REQUEST FOR INFORMATION

United States Space Force Space Systems Command

Space Data Network (SDN) Mission Operations Center (MOC)

1.0 Description

1.1 The USSF's Space System Command is seeking information on fielding a Mission Operations Center (MOC) for Space Data Network (SDN). The Mission Operations Center includes procurement, lifecycle management, technology refreshes, pLEO providers' services management, network performance monitoring, orchestrating space transport network connectivity, user onboarding, information assurance processing, sustainment and other related tasks.

1.2 THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is issued solely for information and planning purposes – it does not constitute a Request for Proposal (RFP) or a promise to issue an RFP in the future. This RFI does not commit the Government to contract for any supply or service whatsoever. Further, the USSF is not at this time seeking proposals and will not accept unsolicited proposals. Respondents are advised that the U.S. Government will not pay for any information or administrative costs incurred in response to this RFI; all costs associated with responding to this RFI will be solely at the interested party's expense. Not responding to this RFI does not preclude participation in any future RFP, if any is issued. Any information submitted by respondents to this RFI is strictly voluntary. All submissions become Government property and will not be returned. Proprietary information will be safeguarded in accordance with the applicable Government regulations. Proprietary information or trade secrets shall be clearly identified.

1.3 NAICS Code: 541715 Aerospace Research and Development (except prototype and production). The small business size standard is 1000. Participation from small and small disadvantaged businesses is highly encouraged.

2.0 Background

2.1 Scope Overview: Space Systems Command (SSC) is seeking information for a contractor to provide a Mission Operations Center for the Space Data Network (SDN). SDN is a resilient network transport that integrates diverse space-based networks and ground infrastructures to deliver private transport, tactical data links, and public internet services to users. The system integrates DoD and commercial pLEO, MEO, and GEO systems to enable packet routing, S-band, and broadband services across the various satellite constellations. The SDN mission operations services will be provided through coordination among six key components: (1) pLEO satellite providers, (2) Ground Entry Point (GEP), (3) Satellite Exchange Point (SEP),

(4) SEP host satellite providers, (5) end-users, and (6) mission partners/external systems. The service assurance, provisioning, operations, and management of SDN services will be performed at the SDN Mission Operations Center. Figure 1 shows a notional SDN topology with the MOC having either direct or indirect network connectivity to other segments/components in system to provide the necessary operational services. Numbered elements represent system components and are described in Table 1.

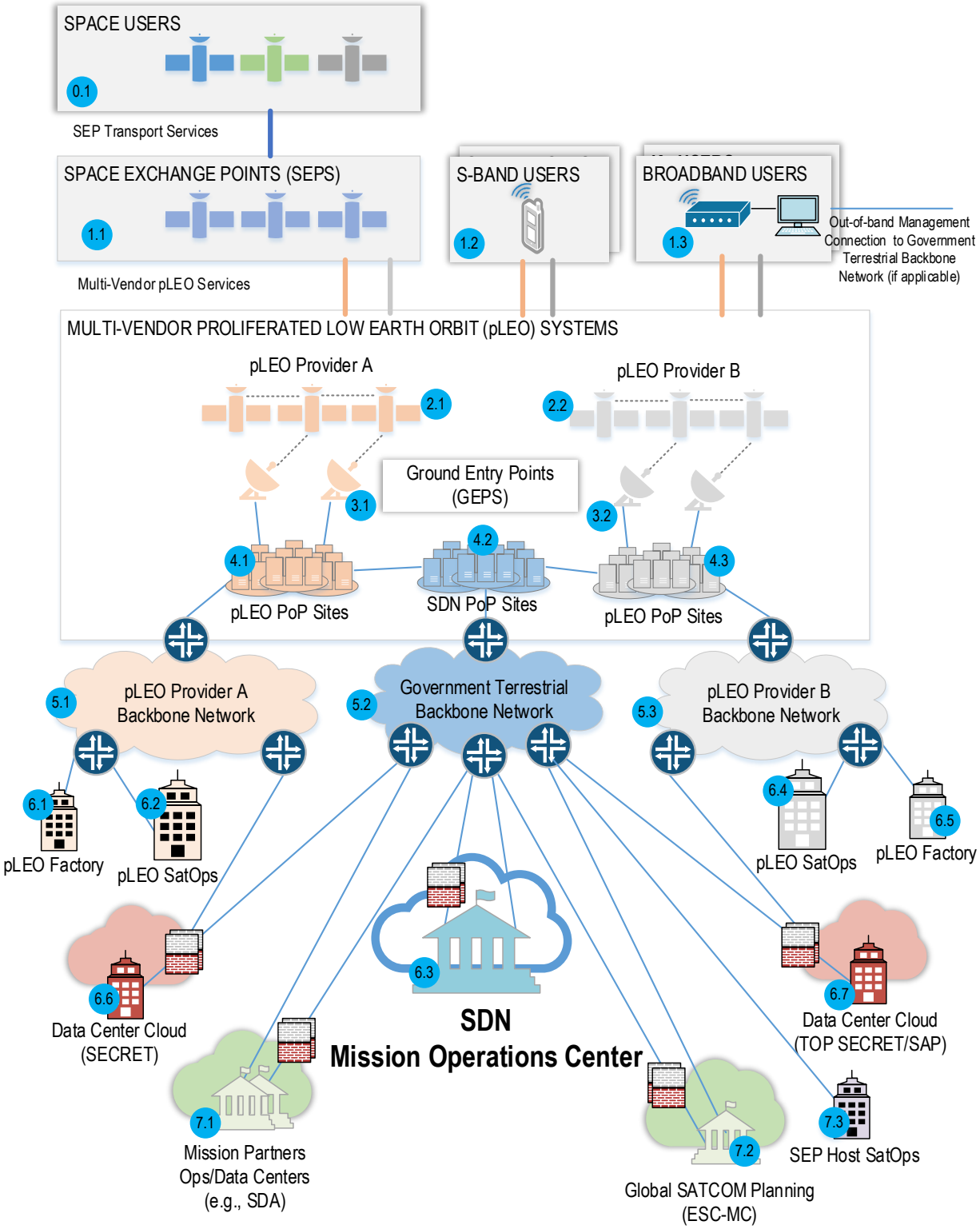

Figure 1. Notional Space Data Network (SDN) Topology

Table 1. System Description

Reference Identifier	System Component	Control/ Manage?	Description
0.1	Space Users	SA-only	LEO/MEO/GEO space assets. SDN transport service subscribers. Serves as the customer edge of the user network.
1.1	Space Exchange Points (SEP)	Yes	A network component of SDN. Serves as the SDN provider edge that connects space users to the SDN backbone; together with edge routers located at the SDN PoP sites, forms the overlay endpoints carrying space users' traffic across the pLEO underlay networks.
1.2	S-Band Users	Partial	Subscribers to the pLEO providers' S-band service, access the services via the S-Band units, which are part of the end-user network.
1.3	Broadband Users	Partial	Subscribers to the pLEO providers' broadband service, access to the service via the Ku/Ka user terminals, which are part of the end-user network.
2.1, 2.2	pLEO Satellites	SA-only	Multi-vendor low-earth orbit satellites providing global coverage that form SDN's underlay transport connectivity.
3.1, 3.2	Ground Exchange Points (GEP)	SA-only	Ground gateway that connects the pLEO satellite constellation to the pLEO point of presence (PoP)
4.1, 4.3	pLEO PoP Sites	SA-only	pLEO provider's point of presence (PoP) is a physical location that hosts network elements used to terminate, aggregate SDN traffic, connecting space networks to terrestrial backbone, interconnects with internet exchange points (IXPs), and customer-facing network edge equipment (i.e., SDN network edge device)
4.2	SDN PoP Sites	Yes	SDN point of presence (PoP) is a physical location that serves as regional handoff point for the SEP user traffic. The facility hosts network elements used to terminate, aggregate SDN user traffic, connecting space users to government terrestrial backbone, and SDN-user-facing network edge equipment. Together with the SEP, it forms the overlay endpoints carrying space users' traffic across the pLEO underlay networks.
5.1, 5.3	pLEO Backbone	SA-only	pLEO provider's terrestrial backbone transport network
5.2	Government Backbone	SA-only	U.S. Government terrestrial backbone transport network
6.1, 6.5	pLEO Factories	SA-only	pLEO provider's factory site
6.2, 6.4	pLEO SatOps	SA-only	pLEO provider's satellite operations center responsible for commanding and processing/storing satellite telemetry
6.3	SDN MOC	Yes	SDN mission operations center. Cloud-based and/or on-premises compute and storage facilities for managing SEP payloads operations, orchestrating the delivery of space transport service, processing user resource requests, resource planning and allocation, and network configurations, fault and performance monitoring, and provisioning of keys/certifications, etc.
6.6, 6.7	Classified Data Center Clouds	Yes	Distributed storage and computing processing centers for classified planning that are part of the MOC
7.1	Mission Partners Ops	SA-only	External mission partners operations and/or processing centers
7.2	Global SATCOM Planning	SA-only	User SATCOM request processing and planning entity
7.3	SEP Host SatOps	SA-only	SEP host satellites operations center

[1]Partial: MOC and the provider of the services have shared control/management of the components. For example: the MOC may control the scheduling and execution of the software upgrades or security rollouts mandated by government policies.

[2]SA-only: MOC receives situational awareness data such as of satellite ephemeris and health/status of the system components from the service/managed providers.

2.2 The MOC will consist of a network operations component that provisions, orchestrates, controls, and manages the network transport services, and maintains the health/status and performance of all network elements that enable these services. The MOC receives and processes services requests from the user segment. The provisioning and orchestration of network services will require coordination and interaction with satellite service providers to plan, allocate and approve the user resource requests. The MOC will also interact with SEPs/GEPs and end-users to execute, deliver and maintain the provisioned transport connectivity throughout the lifetime of the service subscriptions. To assure service performance, the MOC monitors system health/status and performance by directly managing SDN network elements or through ingestion of network logs and situational awareness data from satellite providers. In addition, the MOC provides the front-end services to process service requests from users as well as respond to problems with provisioned services reported by users. *Figure 2* shows a notional diagram illustrating selected real-time and non-real-time messages between logical functional areas of the MOC, as well as the internal messages among those functional areas. The diagram is not intended to exhaustively represent all system messaging. Its purpose is to demonstrate representative message flows that highlight the capabilities and responsibilities of each functional areas.

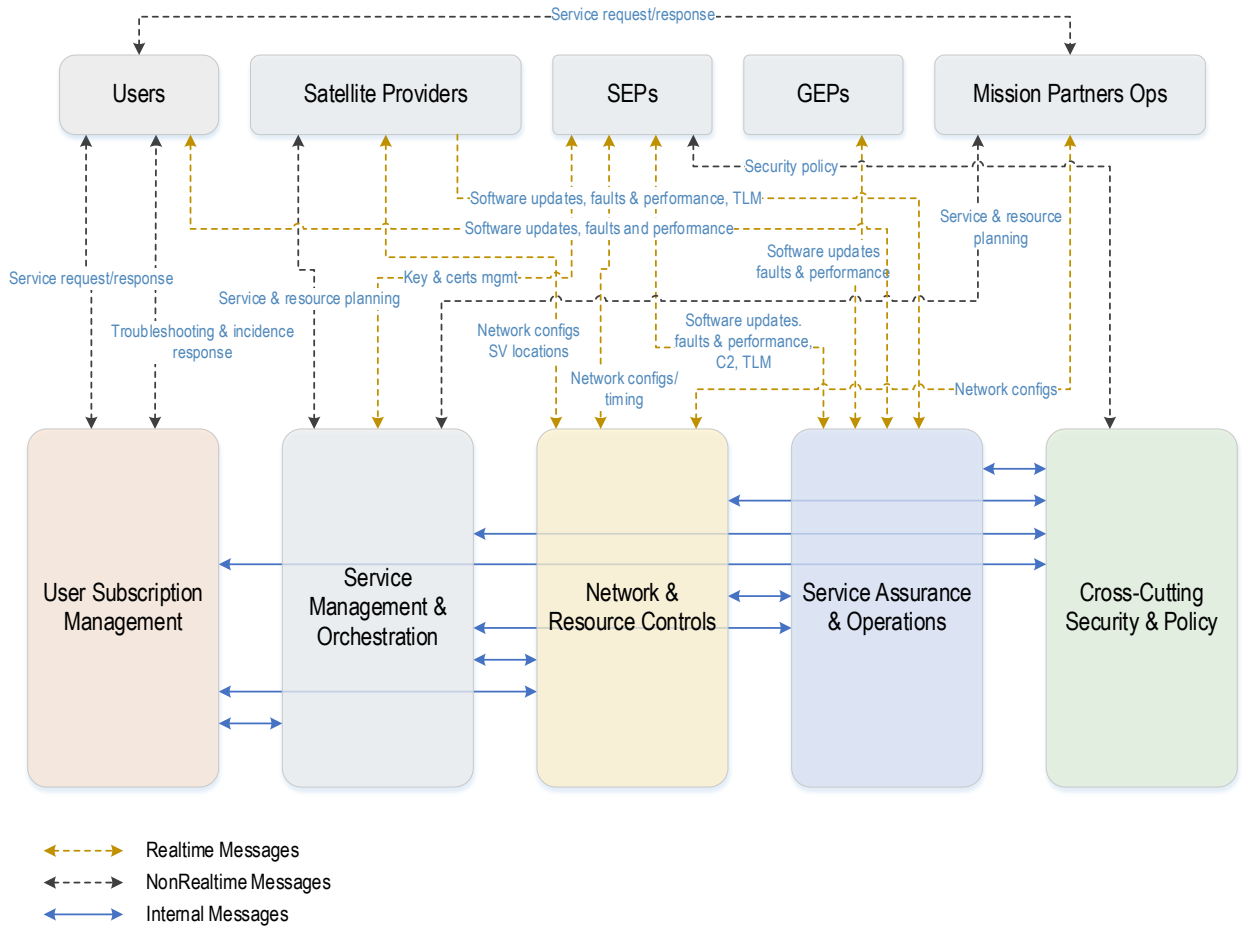

Figure 2. SDN MOC Functional Areas

2.3 Generic Systems to be operated, interfaced with, and/or sustained include but are not limited to:

- 2.3.1 Information technology (IT) infrastructure to communicate at appropriate levels (CLASSIFIED and UNCLASSIFIED)
- 2.3.2 Cybersecurity systems with Risk Management Framework standards
- 2.3.3 Network systems
- 2.3.4 Government systems and transport
- 2.3.5 Government and commercial space systems
- 2.3.6 Commercial satellite providers' satellite operations center
- 2.3.7 Commercial satellite providers' factory operations center
- 2.3.8 Commercial satellite terminals
- 2.3.9 Commercial terrestrial transport

3.0 Assumptions

The physical location for the MOC will be a government provided facility. Day-to-day operations of the system will be performed by contracted personnel with government

oversight. USSF and Combat Forces Command (CFC) will have a mission director in charge of MOC contracted personnel. These individuals will require a government issued CAC for base access as well as access to appropriate networks. Contractors will need to have a minimum TS/SCI clearance.

4.0 MOC Key Systems Objectives:

MOC capabilities will be developed incrementally in phases based on operational timelines for SDN user requirements. Initial capabilities will be developed in the first phase for S-band and broadband users. The second phase will be additive, developing SEP user services to be integrated into the MOC. While developed in phases, the MOC should be designed with this end state in mind.

4.1.1 Phase 1: S Band and Broadband User Services.

4.1.1.1 Provides commercial/government cloud-based and/or government on-premises network and computing infrastructures at multiple security levels with a CIA rating of High/High/Moderate

4.1.1.2 Integrates with new and existing government operational transport networks, government mission user networks, and test networks and applications at different security levels (CUI, Secret, Top Secret and SAP).

4.1.1.3 Provides low-latency cross-domain solutions for processing data across different security levels including CUI, Secret, Top Secret and SAP.

4.1.1.4 Integrates with commercial leased-line providers, Internet service providers, and satellite service providers.

4.1.1.5 Provides redundancy to assure at the minimum 99% operational system availability.

4.1.1.6 Processes both service requests and troubleshooting requests from users.

4.1.1.7 Provides service response and incidence responses to users.

4.1.1.8 On-boards, configures, reconfigures, revokes users and services.

4.1.1.9 Conducts planning, prioritization, and service management to assure services to users.

4.1.1.10 Conducts dynamic capacity planning and resource allocations based on user demands and priority.

4.1.1.11 Monitors S-band, broadband, and network connections to user systems and interfaces.

4.1.1.12 Maintains software and security updates on GEP and user terminals

4.1.1.13 Monitors fault/performance of GEP, and user terminals.

4.1.1.14 Collects and ingests situational awareness data from GEP, pLEO SatOps, pLEO factories, and pLEO PoPs to determine constellation health and status and data trending

4.1.1.15 Provides virtual interfaces to distribute and ingest situational awareness data for local display and cloud-based access.

4.1.1.16 Generates and disseminates faults and performance reports to external segments.

4.1.1.17 Coordinates configurations and software rollout to GEP and user segments.

4.1.1.18 Automates certification rotation, expiration, and revocation.

4.1.1.19 Ensures consistent security policies across layers.

4.1.1.20 Maintains user IP address plans.

4.1.1.21 Provides machine-to-machine and human-to-machine interfaces to add, reconfigure and remove network offerings of the new/existing users.

4.1.1.22 Provides machine-to-machine interface to Application Programming Interface (API) and automation interfaces to government and commercial systems.

4.1.1.23 Provides authentication, authorization, and accounting services.

4.1.1.24 Provides role-based access controls.

4.1.1.25 Provides storage of payload telemetry and network performance data.

4.1.1.26 Complies with industry security and networking standards.

4.1.1.27 Aligns with Modular Open Systems Approach (MOSA) design principles, utilizing open standards, allowing interchangeable modules, independent upgrades and replacements

4.1.2 Phase 2: SEP User Services

4.1.2.1 Monitors SEP users and network connections to user systems and interfaces.

4.1.2.2 Commands and operates SEP payloads through a SEP Network Operations Center (NOC).

4.1.2.3 Provides anomaly resolution of SEP hardware and software components.

4.1.2.4 Orchestrates the provisioning of secured multipoint Layer 2 and Layer 3 virtual private network (VPN) transport services

4.1.2.5 Orchestrates scalable network solution that assigns, and load-balances resources provided by pLEO satellite providers, SEP and GEP segments

4.1.2.6 Orchestrates scalable network solution that provide user traffic separation across SDN.

4.1.2.7 Conducts planning, prioritization, and service management to assure services to users.
4.1.2.8 Provides network topology and route management.
4.1.2.9 Conducts dynamic capacity planning and resource allocations based on user demands and priority.
4.1.2.10 Coordinates software and security updates on SEP, GEP and user terminals.
4.1.2.11 Monitors fault/performance of SEP. Detects, isolates, and corrects faults in the GEP/SEP overlay networks.
4.1.2.12 Provides system timing for the SEP segment.
4.1.2.13 Provides global key management service that spans all layers.
4.1.2.14 Maintains SDN IP address plans.

5.0 Submittal Format:

5.1 Interested parties are requested to respond to this RFI with a white paper and an optional power point slide.

The white paper shall be limited to a single document of no more than 10- pages. A cover letter, if submitted, is excluded from the page limit. The document should be single-spaced, have one-inch margins, US letter-size (8 1/2 by 11 inches) page, and minimum font size of 12-point font.

A PowerPoint slide is allowed to be submitted, in addition to the white paper, but must be limited to a single document of no more than 1 slide. The title slide and information required as part of section 5.2 are excluded from the slide limit. Slide size should be 16:9 aspect ratio.

5.2 Submittal Requirements:
SSC does not intend to award a contract on the basis of responses. Acknowledgement of receipt of responses will be made, but respondents will not be notified of SSC's evaluation of the information received. Responses shall include the following information:

5.2.1 Business name and Size Standard
5.2.2 Submitter's Name and Parent Company, if applicable
5.2.3 Business Address: Street, City, State, and ZIP code
5.2.4 Commercial and Government Entity (CAGE) Code
5.2.5 Unique Entity ID
5.2.6 Point(s) of contact, including telephone number(s) and email address(es)

Please submit PowerPoint presentations and white papers in Microsoft Office compatible form or Portable Document Format (PDF) are due no later than 16:00PT 27 February 2026. Responses shall be via email only to Primary:

melanie.gipson@spaceforce.mil Alternate: brandon.villalpando@spaceforce.mil. Proprietary information, if any, should be minimized and MUST BE CLEARLY MARKED. Please be advised that all submissions become Government property and will not be returned.

SSC has entered into contracts with Federally Funded Research and Development Center (FFRDC) and/or Systems Engineering and Technical Assistance staff working for the Government. These include but are not limited to: The Aerospace Corporation, MIT LL, SAIC, Zachary Piper, SAVI, KBR. These entities may review submissions as technical support to the government. If Respondents disagree to the release of its RFI response to any of the aforementioned firms, the Respondent must clearly state this restriction in the cover letter accompanying the RFI response.

Foreign firms are advised that they will not be allowed to participate in this acquisition at the prime contractor level; however, provided they are eligible to do business with the US Government they may be eligible to participate at the subcontractor level. The research and test data produced under a resultant contract may contain Military Critical Technology List (MCTL) information whose export is restricted by Export Control Act (Title 22, U.S. Sec 2751, et seq.) or the Export Administration Act of 1979, as amended (Title 50, U.S.C., App. 2401, et seq.). Request certification and registration from the Defense Logistics Services Center (DLSC), Federal Center, 74 North Washington, Battle Creek, MI 49016-3412 as soon as possible.

5.3 Responses

SSC is seeking responses related to vendors' products and service offerings. Vendors' responses should provide information about their capability for performing the SDN MOC's objectives as outlined in section 4.0 and elaborate on any requirements. SSC understands that there may be other pertinent information that has not been solicited and encourages vendors to provide additional information as deemed appropriate.

SSC is seeking information on potential MOC design and implementation approaches for the Space Data Network (SDN) that can readily interface with existing government and commercial pLEO systems, U.S. government transport networks, and future SEP systems, to enable rapid deployment of assured mission services to end-user systems. The RFI aims to gather vendor's partial or full scope capability of a cloud-based and/or on-premises mission operations center that provides services not only for managing user service subscriptions from multiple U.S. government agencies but also planning and managing resource allocation and service performance from multiple satellite vendors, as well as for interfacing with external mission partners to

extend the data network transport capability across the enterprise. SSC welcomes insights into technology roadmaps, comprehensive satellites and network resource planning and orchestration methodologies that account for diverse services and user system types in a multi-vendors environment, and network observability approaches that scale to support multiuser integrated system while meeting performance and resilience requirements. Respondents should describe how the proposed solution enables incremental capability expansion without requiring redesign or replacement of core components. Participants are encouraged to provide clear documentation, relevant case studies, and innovative approaches that meet our outlined requirements..

This RFI process seeks responses from any and all respondents who offer partial or total solutions in order to assess methodologies and develop a range of total costs association with each solution.

Respondents shall provide detailed information on the following topic areas. Each topic and sub-topic shall have a separate response.

- 5.3.1 Design and implementation of SDN mission operations center that performs the objectives outlined in section 4.0.
  - 5.3.1.1 Describe scalability of the proposed system and how it will support multi-providers solutions and globally distributed GEPs, PoPs, and user terminals.
  - 5.3.1.2 Describe open standards and protocols that pertain to orchestrating, managing, and operating a space-based transport that utilizes software defined network technologies.
  - 5.3.1.3 Describe a solution that supports incremental expansion (e.g., addition of pLEO providers or new mission areas) without requiring redesign of the core capabilities.
  - 5.3.1.4 Describe how the proposed architecture enables future integration with external systems, tools, and data sources without requiring redesign of the core MOC platform.
  - 5.3.1.5 Describe your software architecture and design approach, including computing requirements, and percentage of commercial-of-the-shelf products versus development required to meet the MOC objectives. For the COTS components, please identify the specific products/platforms.
  - 5.3.1.6 Describe how your proposed solution synchronizes and distributes time to SEPs.
  - 5.3.1.7 Describe how the proposed solution meets user quality of service/quality of experience in a system that integrates multiple service providers.

5.3.1.8 Explain your strategy for orchestrating diverse and redundant network connectivity across various satellites providers to assure user service availability.

5.3.1.9 Describe your strategy in utilizing network observability tools to correlate issues across multi-vendor network services, hardware, and integrated systems to gain insight into system performance to achieve a faster anomaly resolution.

5.3.1.10 Provide an overview of a helpdesk, engineering, maintenance, and operational support for a multi-security level SDN MOC.

5.3.2 Identification and assessment of key system design trades

5.3.2.1 Cloud versus on-premises MOC.

5.3.2.2 MOC-managed versus network provider-managed SDN PoPs.

5.3.2.3 Identification of cost drivers in the design and implementation.

5.3.2.4 Assessment of SDN MOC information assurance architecture.

5.3.2.5 Security and threat prevention capabilities

5.3.3 Assessment of development and production schedules.

5.3.4 Estimation of development and operational costs.

5.3.5 Describe the extent your solution meets the commercial definition in FAR 2.101.

5.3.6 List of questions to help clarify the scope of the MOC.

5.4 THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. The information provided in the RFI is subject to change and is not binding on the Government. The USSF has not made a commitment to procure any of the items discussed, and release of this RFI should not be construed as such a commitment or as authorization to incur cost for which reimbursement would be required or sought. All submissions become Government property and will not be returned.